Mixed-current external-photon QCD sum rules for $D_{s1}, B_{s1} \to D_s^*, B_s^*\gamma$

Clean-room symbolic and numerical analysis

29 July 2026

Abstract

We construct a 2×2 QCD sum-rule matrix for the axial-vector and derivative-tensor heavy-light currents, rotate it to lower and higher 1^+ channels, and combine the resulting residues with an external-photon light-cone sum rule (LCSR). The transition operator product expansion (OPE) retains leading-order hard emission and Ball–Braun–Kivel two- and three-particle photon distribution amplitudes (DAs) through twist four. Ordinary local vacuum-condensate transition terms are excluded; the historical local heavy-line term is displayed only as a diagnostic. The tensor-current transition is derived at leading power, where $\widehat{T}_B = m_Q \widehat{T}_A/(m_Q + m_s)$ after the double Borel transform. Independent Python and Wolfram Language implementations agree for 42/42 regression entries, with maximum material relative difference 9.13e-07. From 1200 accepted uncertainty samples per state we obtain median radiative widths of 0.482, 0.094, 1.589, and 2.332 keV for $D_{s1}(2460)$, $D_{s1}(2536)$, a predicted lower B_{s1}^L, and $B_{s1}(5830)$, respectively. The distributions are broad and non-Gaussian. In particular, the transition Borel-mass variation is large, so these results are exploratory rather than precision predictions.

1 Scope and applicability

The channels studied are

$$\begin{aligned} D_{s1}(2460)^+ &\to D_s^{*+}\gamma, & D_{s1}(2536)^+ &\to D_s^{*+}\gamma, \\ B_{s1}^{L\,0} &\to B_s^{*0}\gamma, & B_{s1}(5830)^0 &\to B_s^{*0}\gamma. \end{aligned} \tag{1}$$

They are heavy-light mesons with one strange spectator and an on-shell photon. An external-photon LCSR is therefore applicable. The lower bottom state is a model prediction, whereas $B_{s1}(5830)$ is established. The particle masses and experimental status are taken from the 2026 Particle Data Group listings [1]. The photon DAs and their normalization follow Ref. [2]; the pure-axial charm and bottom analyses used as benchmarks are Refs. [3, 4].

This calculation has a sharply declared boundary. It is leading order in α_s and includes transition DAs through twist four. The full two-point AA, AB, and BB matrix is calculated. For the transition only the leading-power tensor projection is retained. Subleading tensor kernels and transition radiative corrections are not silently estimated as calculated terms: correlated truncation uncertainties of 15% in charm and 5% in bottom are assigned to the tensor projection, and both omissions remain explicit in the validation ledger.

2 Conventions, currents, and observable

We use $g_{\mu\nu} = \mathrm{diag}(1, -1, -1, -1)$, $\epsilon^{0123} = +1$, $\gamma_5 = i\gamma^0\gamma^1\gamma^2\gamma^3$, and $\sigma_{\mu\nu} = i[\gamma_\mu, \gamma_\nu]/2$. The initial momentum is $p' = p + q$, the photon satisfies $q^2 = 0$, and $e > 0$. The currents are

$$J_A^\mu = \bar{s}\gamma^\mu\gamma_5 Q, \qquad J_B^\mu = \frac{i}{m_Q + m_s}\bar{s}\sigma^{\mu\nu}p'_\nu\gamma_5 Q, \qquad J_V^\mu = \bar{s}\gamma^\mu Q. \tag{2}$$

The physical-current rotation is

$$\begin{pmatrix} J_L \\ J_H \end{pmatrix} = R(\theta) \begin{pmatrix} J_A \\ J_B \end{pmatrix}, \qquad R(\theta) = \begin{pmatrix} \sin\theta & \cos\theta \\ \cos\theta & -\sin\theta \end{pmatrix}. \tag{3}$$

The adopted angles are $\theta_D = 38.4^\circ$ and $\theta_B = 35.264^\circ$, with angle scans included below. The pole residues are defined without hidden current-dependent factors,

$$\langle 0|J_i^\mu|H_i(p', \eta)\rangle = f_i m_i \eta^\mu, \qquad \langle 0|J_V^\mu|V(p, \xi)\rangle = f_V m_V \xi^\mu. \tag{4}$$

For an on-shell $1^+ \to 1^-\gamma$ transition there are two independent gauge-invariant multipoles. We calculate the leading antisymmetric structure

$$\mathcal{A} = ie\, g\, \epsilon_{\alpha\beta\rho\sigma} \eta^\alpha \xi^{*\beta} \varepsilon^{*\rho} q^\sigma, \tag{5}$$

and set the second, higher-multipole form factor to zero at the retained leading-power accuracy. Replacing ε by q makes equation (5) vanish identically. The corresponding width is

$$\Gamma(1^+ \to 1^-\gamma) = \frac{\alpha_{\rm em} g^2 k_\gamma^3 (m_i^2 + m_f^2)}{3 m_i^2 m_f^2}, \qquad k_\gamma = \frac{m_i^2 - m_f^2}{2m_i}. \tag{6}$$

Thus g is dimensionless.

3 Two-point matrix sum rule

3.1 Cut trace and exact-mass spectral densities

Define

$$\Pi^{ij}_{\mu\nu}(p') = i \int d^4x\, e^{ip'\cdot x} \langle 0|T\{J^i_\mu(x) J^{j\dagger}_\nu(0)\}|0\rangle = (-g_{\mu\nu} + p'_\mu p'_\nu / p'^2)\Pi^{\rm T}_{ij} + \cdots. \tag{7}$$

At leading order, the cut loop contains a heavy quark and an anti-strange quark. In the p' rest frame, the transverse spin average is

$$\frac{1}{3}\sum_{a=1}^{3} {\rm Tr}[(\not{\ell} - m_s)\Gamma_i^a(\not{k} + m_Q)\bar{\Gamma}_j^a], \quad \Gamma_A^a = \gamma^a\gamma_5, \quad \Gamma_B^a = \frac{i\sqrt{s}}{m_Q + m_s}\sigma^{a0}\gamma_5. \tag{8}$$

The two-body phase-space cut gives the exact-m_s spectral matrix. With $S = m_Q + m_s$, $d = m_Q - m_s$, and

$$\lambda(s, m_Q^2, m_s^2) = [s - S^2][s - d^2], \tag{9}$$

the entries are

$$\rho_{AA}(s) = \frac{\sqrt{\lambda}}{8\pi^2}\frac{(s - S^2)(d^2 + 2s)}{s^2}, \tag{10}$$

$$\rho_{AB}(s) = \rho_{BA}(s) = \frac{\sqrt{\lambda}}{8\pi^2}\frac{3d(s - S^2)}{Ss}, \tag{11}$$

$$\rho_{BB}(s) = \frac{\sqrt{\lambda}}{8\pi^2}\frac{(s - S^2)(s + 2d^2)}{S^2 s}. \tag{12}$$

All entries have dimension two and begin at the physical threshold S^2. The symbolic determinant factorizes as

$$\det\rho(s) = \frac{(s - S^2)^3 (s - d^2)^3}{32\pi^4 S^2 s^3} \geq 0 \qquad (s \geq S^2), \tag{13}$$

which checks spectral positivity as well as the AB normalization.

3.2 Pre-Borel representation and local terms

Before Borel transformation,

$$\Pi^{\mathrm{T}}_{ij}(p'^2) = \int_{S^2}^{\infty} ds\, \frac{\rho_{ij}(s)}{s-p'^2} + \Pi^{(3)}_{ij}(p'^2) + \Pi^{(m_s)}_{ij}(p'^2) + \Pi^{(5)}_{ij}(p'^2). \tag{14}$$

The Borel rule used everywhere is

$$\mathcal{B}_{P^2\to M^2}\frac{1}{(s-P^2)^n} = \frac{e^{-s/M^2}}{(n-1)!(M^2)^{n-1}}. \tag{15}$$

Writing $\langle\bar{s}s\rangle = s_s$ and $\langle\bar{s}g_s\sigma Gs\rangle = m_0^2 s_s$, the local Borel matrices are

$$\widehat{T}\Pi^{(3)} = s_s e^{-m_Q^2/M^2}\begin{pmatrix} m_Q & m_Q^2/S \\ m_Q^2/S & m_Q^3/S^2 \end{pmatrix}, \tag{16}$$

$$\widehat{T}\Pi^{(m_s)} = s_s e^{-m_Q^2/M^2}\left(-\frac{m_Q^2 m_s}{2M^2} + \frac{m_Q^3 m_s^2}{2M^4}\right)\begin{pmatrix} 1 & m_Q/S \\ m_Q/S & m_Q^2/S^2 \end{pmatrix}, \tag{17}$$

$$\widehat{T}\Pi^{(5)} = e^{-m_Q^2/M^2}\begin{pmatrix} -\frac{m_0^2 s_s m_Q^3}{4M^4} & -\frac{m_0^2 s_s}{4S}\left(\frac{m_Q^4}{M^4} - \frac{m_Q^2}{M^2} - 1\right) \\ \text{sym.} & \frac{m_0^2 s_s}{S^2}\left(-\frac{m_Q^5}{4M^4} + \frac{m_Q^3}{2M^2}\right) \end{pmatrix}. \tag{18}$$

The continuum-subtracted basis matrix is therefore

$$\widehat{T}\Pi_{ij}(M^2, s_0) = \int_{S^2}^{s_0} ds\, e^{-s/M^2}\rho_{ij}(s) + \widehat{T}\Pi^{(3)}_{ij} + \widehat{T}\Pi^{(m_s)}_{ij} + \widehat{T}\Pi^{(5)}_{ij}. \tag{19}$$

No off-diagonal element is set to zero. The physical matrix is

$$\widehat{T}\Pi^{\mathrm{phys}} = R(\theta)\widehat{T}\Pi R^T(\theta). \tag{20}$$

For state $a = L, H$,

$$f_a(M^2, s_0) = \frac{e^{m_a^2/(2M^2)}}{m_a}\sqrt{\widehat{T}\Pi^{\mathrm{phys}}_{aa}}, \qquad m^2_{\mathrm{SR}} = \frac{-\partial_{(1/M^2)}\widehat{T}\Pi^{\mathrm{phys}}_{aa}}{\widehat{T}\Pi^{\mathrm{phys}}_{aa}}. \tag{21}$$

The remaining rotated off-diagonal correlation is reported as a diagnostic, not erased by hand.

For the vector final state, the perturbative density is

$$\rho_V(s) = \frac{\sqrt{\lambda}}{8\pi^2}\left[2 - \frac{m_Q^2 + m_s^2 - 6m_Q m_s}{s} - \frac{(m_Q^2 - m_s^2)^2}{s^2}\right], \tag{22}$$

and the local AA coefficients in equations (16) to (18) change sign. Its residue is extracted with the analogue of equation (21).

3.3 Window criteria

The pole contribution is the continuum-subtracted correlator divided by the same OPE with the perturbative upper limit extended to infinity. Accepted two-point points obey

$$\mathrm{PC} \geq 0.40, \qquad \frac{|\Pi^{(5)}|}{|\Pi|} \leq 0.10, \qquad \left|\frac{m_{\mathrm{SR}}}{m_{\mathrm{phys}}} - 1\right| \leq 0.05. \tag{23}$$

The resulting windows and diagnostics are:

State	M^2 [GeV^2]	s_0 [GeV^2]	accepted	PC	$\lvert d=5\rvert/\lvert\Pi\rvert$	m_{SR} [GeV]
$D_{s1}(2460)$	2.0–3.0	7.50–8.50	29/30	0.41–0.71	0.023	2.418–2.506
$D_{s1}(2536)$	2.0–3.0	8.50–10.00	18/30	0.41–0.67	0.097	2.428–2.655
B^L_{s1}	8.0–11.0	40.00–42.00	30/30	0.42–0.66	0.002	5.649–5.785
$B_{s1}(5830)$	7.0–9.0	42.00–44.00	30/30	0.40–0.64	0.037	5.730–5.896

Table 1: Accepted two-point windows. The predicted lower bottom mass used for the residue extraction is 5.750 GeV.

4 External-photon transition sum rule

4.1 Correlation function, contraction, and OPE routing

For basis current $i = A, B$,

$$T^i_{\mu\nu}(p,q) = i\int d^4x\, e^{ip\cdot x}\langle\gamma(q,\varepsilon)|T\{J^\dagger_{V\mu}(x)J^i_\nu(0)\}|0\rangle. \tag{24}$$

The Wick contraction is fixed as

$$T^i_{\mu\nu} = -i\int d^4x\, e^{ip\cdot x}\langle\gamma|\bar{s}(x)\gamma_\mu S_Q(x,0)\Gamma^i_\nu s(0)|0\rangle, \tag{25}$$

which fixes the global fermion sign. The light-cone propagators used to generate the nonlocal terms include

$$S^{(0)}_s(x) = \frac{i\not{x}}{2\pi^2x^4} - \frac{m_s}{4\pi^2x^2}, \tag{26}$$

$$S^{(\gamma)}_Q(x,0) = -ie_Q\int d^4y\, S^{(0)}_Q(x-y)\not{A}(y)S^{(0)}_Q(y), \tag{27}$$

$$S^{(G)}_Q(k) = -\frac{ig_s}{4}\frac{G_{\alpha\beta}\left[\sigma^{\alpha\beta}(\not{k}+m_Q)+(\not{k}+m_Q)\sigma^{\alpha\beta}\right]}{(k^2-m_Q^2)^2}. \tag{28}$$

For the nonlocal strange propagator we retain the free part and the one-gluon light-cone insertion. Ordinary vacuum-condensate pieces are not inserted into the transition correlator. This prevents a local $\langle\bar{s}s\rangle$ term from being mixed with the external-photon DA sum rule.

The needed two-particle photon matrix elements are, schematically,

$$\langle\gamma|\bar{s}(x)\sigma_{\alpha\beta}s(0)|0\rangle = -ie_s s_s(\varepsilon_\alpha q_\beta - \varepsilon_\beta q_\alpha)\int_0^1 du\, e^{iuq\cdot x}\left[\chi\phi_\gamma(u) + \frac{x^2}{16}\mathcal{A}(u)\right] + \cdots, \tag{29}$$

$$\langle\gamma|\bar{s}(x)\gamma_\alpha s(0)|0\rangle = e_s f_{3\gamma}\left(\varepsilon_\alpha - q_\alpha\frac{\varepsilon\cdot x}{q\cdot x}\right)\int_0^1 du\, e^{iuq\cdot x}\psi^v(u), \tag{30}$$

$$\langle\gamma|\bar{s}(x)\gamma_\alpha\gamma_5 s(0)|0\rangle = -\frac{e_s f_{3\gamma}}{4}\epsilon_{\alpha\beta\rho\sigma}\varepsilon^\beta q^\rho x^\sigma\int_0^1 du\, e^{iuq\cdot x}\psi^a(u). \tag{31}$$

The three-particle set $\{\mathcal{S}, \widetilde{\mathcal{S}}, \mathcal{A}, \mathcal{V}, \mathcal{T}_{1,2,3,4}\}$ follows Ref. [2]. Their explicit conformal forms and the combinations entering this projection are listed in section A.

4.2 Pre-Borel kernels

Let

$$D(u) = m_Q^2 - (1-u)p^2 - up'^2, \quad u_0 = \frac{M_1^2}{M_1^2+M_2^2}, \quad \mathcal{M}^2 = \frac{M_1^2M_2^2}{M_1^2+M_2^2}. \tag{32}$$

The double-Borel identity is

$$\mathcal{B}_{M_1^2}\mathcal{B}_{M_2^2}\frac{1}{D(u)^n} = \frac{(\mathcal{M}^2)^{2-n}}{\Gamma(n)}e^{-m_Q^2/\mathcal{M}^2}\delta(u-u_0). \tag{33}$$

We take $M_1^2 = M_2^2 = 2M^2$, so $u_0 = 1/2$ and $\mathcal{M}^2 = M^2$. Up to the common Lorentz structure, the pre-Borel ledger is

$$T_{\text{hard}} = \int_{S^2}^\infty ds\, \frac{\rho_T(s)}{(s-p^2)(s-p'^2)}, \tag{34}$$

$$T_{\text{tw2}} = e_s m_Q s_s \chi\int_0^1 du\, \frac{\phi_\gamma(u)}{D(u)}, \tag{35}$$

$$T_{\rm tw4,2p} = e_s m_Q s_s \int_0^1 du \left[-\frac{m_Q^2 \mathcal{A}(u)}{2D(u)^3} - \frac{(1-u)H_\gamma(u) + \bar{H}_\gamma(u)}{D(u)^2} + \frac{4m_Q^2 \bar{H}_\gamma(u)}{D(u)^3} \right], \tag{36}$$

$$T_{\rm tw3,2p} = e_s f_{3\gamma} \int_0^1 du \left[\frac{C_1(u)}{D(u)} - \frac{2m_Q^2 \Psi^v(u)}{D(u)^2} \right], \tag{37}$$

where

$$C_1(u) = \frac{(1-u)\psi^{a\,\prime}(u)}{4} - \frac{\psi^a(u)}{4} - \Psi^v(u) + (1-u)\psi^v(u), \tag{38}$$

$H_\gamma(u) = \int_0^u h_\gamma(v)dv$, $\bar{H}_\gamma(u) = \int_0^u (u-v)h_\gamma(v)dv$, and $\Psi^v(u) = \int_0^u \psi^v(v)dv$. The three-particle twist-four kernel is proportional to $D^{-2}\mathcal{F}_2$, and the twist-three kernel to $D^{-1}\mathcal{F}_3$, with their exact integration domains in section A.

The hard spectral density, including both quark charges and exact m_s, is

$$\rho_T(s) = \frac{3m_s m_Q}{4\pi^2} \left[e_s \ln \frac{a_s - \sqrt{\lambda}}{a_s + \sqrt{\lambda}} + e_Q \ln \frac{a_Q - \sqrt{\lambda}}{a_Q + \sqrt{\lambda}} \right], \tag{39}$$

$$a_s = s - m_Q^2 + m_s^2, \qquad a_Q = s - m_s^2 + m_Q^2. \tag{40}$$

Changing c to b therefore changes both m_Q and e_Q.

4.3 Post-Borel ledger and physical projection

Define $E = e^{-m_Q^2/M^2}$, $E_0 = e^{-s_0/M^2}$, and evaluate all DAs at u_0. The continuum-subtracted terms are

$$\widehat{T}T_{\rm hard} = \int_{S^2}^{s_0} ds\, e^{-s/M^2} \rho_T(s), \tag{41}$$

$$\widehat{T}T_{\rm tw2} = e_s m_Q s_s \chi M^2 (E - E_0)\phi_\gamma, \tag{42}$$

$$\widehat{T}T_{\rm tw4,2p} = e_s m_Q s_s E \left[-\frac{m_Q^2}{4M^2}\mathcal{A} - (1-u_0)H_\gamma - \bar{H}_\gamma \left(\frac{2m_Q^2}{M^2} \right) \bar{H}_\gamma \right], \tag{43}$$

$$\widehat{T}T_{\rm tw3,2p} = e_s f_{3\gamma} M^2 (E - E_0) \left[\frac{(1-u_0)\psi^{a\,\prime}}{4} - \frac{\psi^a}{4} - (1 + 2m_Q^2/M^2)\Psi^v + (1-u_0)\psi^v \right], \tag{44}$$

$$\widehat{T}T_{\rm tw4,3p} = m_Q e_s s_s E\, I_{\mathcal{F}_2}, \tag{45}$$

$$\widehat{T}T_{\rm tw3,3p} = -e_s f_{3\gamma} M^2 (E - E_0) I_{\mathcal{F}_3}. \tag{46}$$

The quoted axial-basis invariant is the sum of precisely these six entries. For comparison only, the historical local term

$$\widehat{T}T_{\rm local,diag} = e_Q m_Q s_s E \left[1 - \frac{m_s^2}{M^2} \left(1 - \frac{m_Q^2}{M^2} \right) \right] \tag{47}$$

is printed in every term ledger, but it is excluded from all quoted couplings.

For the tensor current the twist-two trace ratio before Borel transformation is

$$\frac{T_B^{(2)}}{T_A^{(2)}} = \frac{k^2}{m_Q(m_Q + m_s)}. \tag{48}$$

Using $k^2/(m_Q^2 - k^2) = m_Q^2/(m_Q^2 - k^2) - 1$, the contact term has zero Borel image and the retained leading-power relation becomes

$$\widehat{T}T_B = \frac{m_Q}{m_Q + m_s} \widehat{T}T_A. \tag{49}$$

We apply equation (49) term by term through twist four. This is the stated leading-power truncation, not a claim that all subleading tensor-current kernels have been calculated.

Let r_{aA} and r_{aB} be row a of $R(\theta)$. The physical invariant and coupling are

$$\widehat{T}T_a = r_{aA} \widehat{T}T_A + r_{aB} \widehat{T}T_B, \tag{50}$$

$$g_a(M^2,s_0)=\frac{e^{(m_a^2+m_V^2)/(2M^2)}}{f_af_Vm_am_V}\widehat{T}T_a. \tag{51}$$

The separate A and B components are retained in the output to expose constructive or destructive interference.

5 Inputs and uncertainty protocol

The fixed-scale LO inputs are $m_c = 1.27(2)$ GeV, $m_b = 4.18(3)$ GeV, $m_s = 9.3(11)\times10^{-2}$ GeV, $\langle\bar{q}q\rangle = -(0.24\pm0.01)^3\text{GeV}^3$, $\langle\bar{s}s\rangle/\langle\bar{q}q\rangle = 0.8\pm0.1$, $m_0^2 = 0.8\pm0.1\text{GeV}^2$, $\chi = -3.15\pm0.30\text{GeV}^{-2}$, and $f_{3\gamma} = -(3.9\pm2.0)\times10^{-3}\text{GeV}^2$ at $\mu = 1$ GeV. The DA shape parameters are $\omega_\gamma^V = 3.8\pm1.8$, $\omega_\gamma^A = -2.1\pm1.0$, $\kappa = 0.2\pm0.2$, $\zeta_1 = 0.4\pm0.2$, and $\zeta_2 = 0.3\pm0.2$. The complete source, unit, scale, status, and note for every input appears in `inputs/input_provenance.csv`.

We use 1200 reproducible samples per state (seed 20260729). External inputs, condensates, DA parameters, mixing angles, Borel variables, and thresholds are varied simultaneously. Accepted two-point points must satisfy the gates above. The tensor truncation nuisance is correlated across transition terms within a sample. The reported central value is the median and the interval is the 16th–84th percentile range. Because g may cross zero, the width distribution is strongly non-Gaussian; consequently a width constructed by squaring the median g need not equal the median width.

6 Results

6.1 Residues, couplings, and widths

State	g	Γ_γ [keV]	f_i [GeV]	f_V [GeV]
$D_{s1}(2460)$	$-0.061^{+0.174}_{-0.160}$	$0.482^{+1.609}_{-0.434}$	$0.404^{+0.020}_{-0.018}$	$0.253^{+0.008}_{-0.010}$
$D_{s1}(2536)$	$-0.017^{+0.058}_{-0.069}$	$0.094^{+0.520}_{-0.089}$	$0.166^{+0.014}_{-0.012}$	$0.253^{+0.008}_{-0.010}$
B_{s1}^L	$-0.548^{+0.320}_{-0.339}$	$1.589^{+2.703}_{-1.298}$	$0.501^{+0.018}_{-0.017}$	$0.300^{+0.009}_{-0.009}$
$B_{s1}(5830)$	$-0.484^{+0.286}_{-0.425}$	$2.332^{+5.735}_{-1.925}$	$0.098^{+0.014}_{-0.008}$	$0.300^{+0.009}_{-0.009}$

Table 2: Monte Carlo medians and 16th–84th percentile intervals. The sign of g follows equation (5); the width is sign independent.

For $D_{s1}(2460)$ the interval in g crosses zero; its width distribution therefore has a long upper tail. The higher charm state has a destructive A/B interference and an even smaller median coupling. The bottom medians are larger in magnitude, but the transition Borel dependence is also more severe. Using the measured total widths where available gives:

State	central radiative branching fraction	experimental status
$D_{s1}(2536)$	1.019×10^{-4}	possibly seen; no branching fraction
$B_{s1}(5830)$	4.664×10^{-3}	mode not observed

Table 3: Derived central branching fractions. The $B_{s1}(5830)$ entry uses the PDG central total width and is therefore especially uncertain.

The central OPE ledger, in GeV^4, is shown in table 4. The last column is excluded from the quoted total.

State	hard	tw2	tw3(2p)	tw3(3p)	tw4(2p)	tw4(3p)	local diag.
$D_{s1}(2460)$	1.900×10^{-2}	-5.100×10^{-2}	1.300×10^{-2}	1.300×10^{-2}	-8.631×10^{-4}	-6.133×10^{-21}	-7.000×10^{-3}
$D_{s1}(2536)$	2.000×10^{-2}	-5.400×10^{-2}	1.400×10^{-2}	1.400×10^{-2}	-8.631×10^{-4}	-6.133×10^{-21}	-7.000×10^{-3}
B_{s1}^L	1.300×10^{-2}	-2.300×10^{-2}	2.000×10^{-3}	2.000×10^{-3}	-4.000×10^{-3}	-1.515×10^{-21}	8.400×10^{-4}
$B_{s1}(5830)$	1.300×10^{-2}	-2.300×10^{-2}	2.000×10^{-3}	2.000×10^{-3}	-4.000×10^{-3}	-1.515×10^{-21}	8.400×10^{-4}

Table 4: Transition term decomposition at the central (M^2, s_0) point. "2p" and "3p" denote two- and three-particle photon DAs.

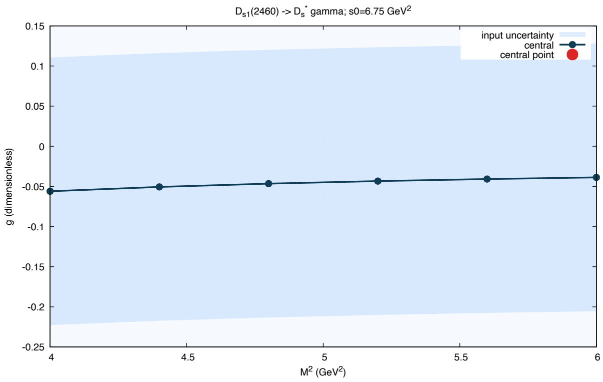
6.2 Stability

The relative full-range variations are:

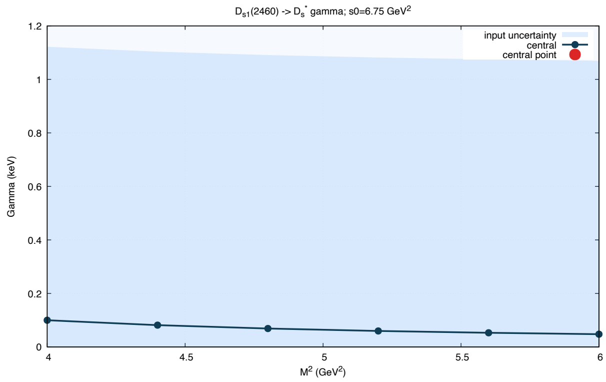
State	$\Delta_{M^2} g$	$\Delta_{s_0} g$	$\Delta_{M^2} \Gamma$	$\Delta_{s_0} \Gamma$
$D_{s1}(2460)$	39.8%	3.4%	87.1%	6.7%
$D_{s1}(2536)$	40.7%	3.3%	89.3%	6.6%
B_{s1}^L	108.6%	1.2%	279.9%	2.3%
$B_{s1}(5830)$	113.4%	0.9%	295.7%	1.9%

Table 5: Relative scan variation $(\max - \min)/|\text{mean}|$.

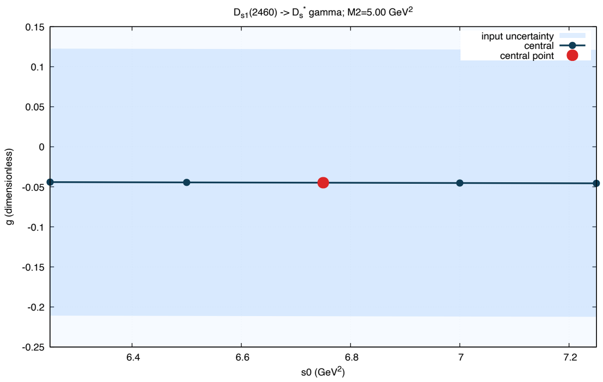
Threshold sensitivity is small on the declared grids, but the Borel-mass sensitivity is not. The coupling changes by about 40% in charm and more than 100% in bottom, while the width variation is larger still. The uncertainty intervals in table 2 reflect this behavior. No plateau claim is made for the bottom transition.



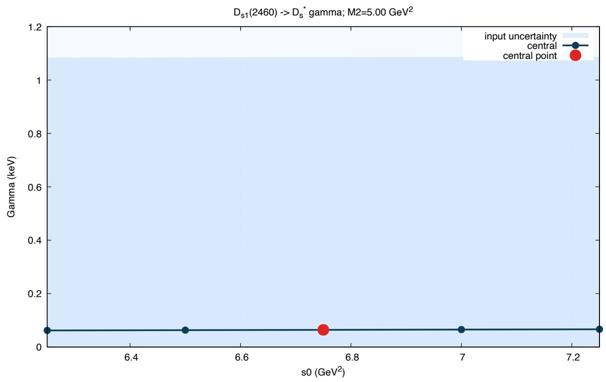
(a) $D_{s1}(2460)$: g vs. M^2.



(b) $D_{s1}(2460)$: width vs. M^2.

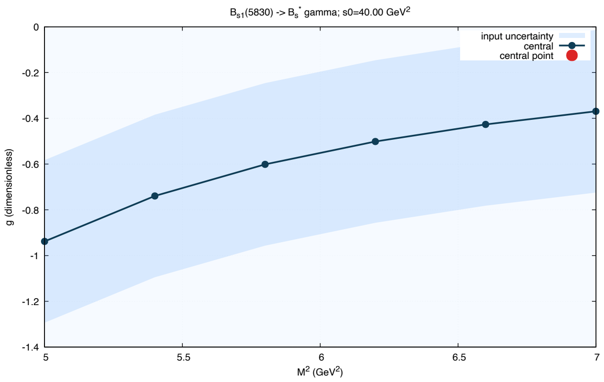


(c) $D_{s1}(2460)$: g vs. s_0.

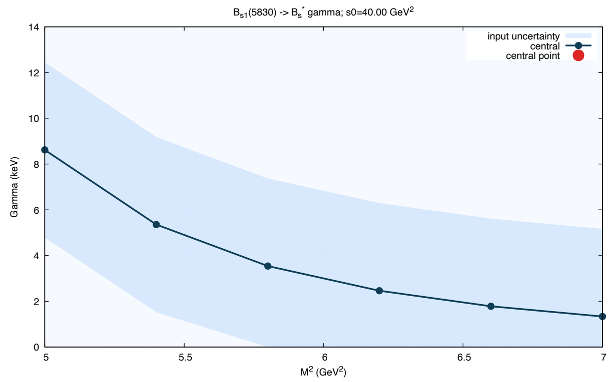


(d) $D_{s1}(2460)$: width vs. s_0.

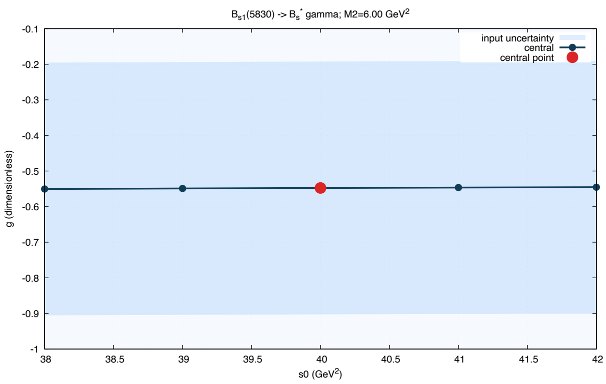
Figure 1: The four principal stability scans for the lower charm state.



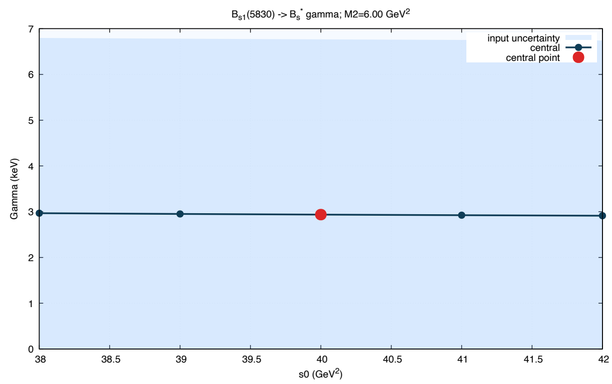
(a) $B_{s1}(5830)$: g vs. M^2.



(b) $B_{s1}(5830)$: width vs. M^2.

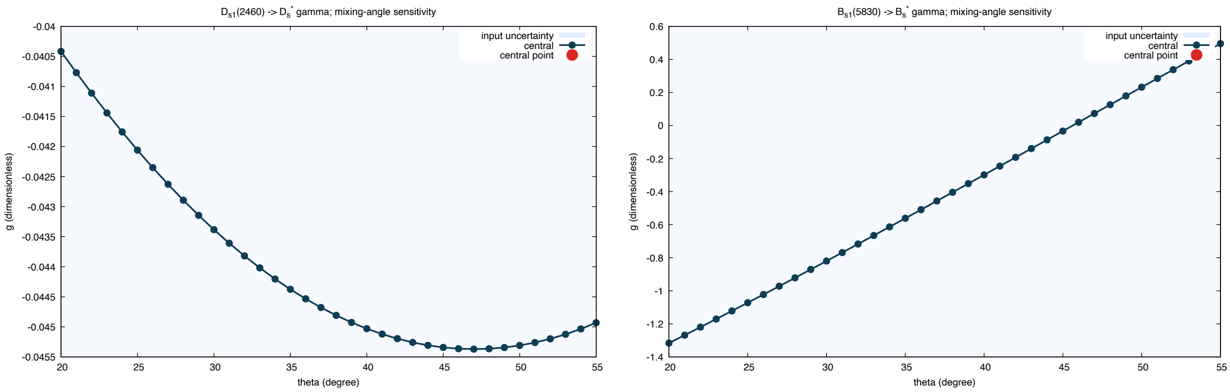


(c) $B_{s1}(5830)$: g vs. s_0.



(d) $B_{s1}(5830)$: width vs. s_0.

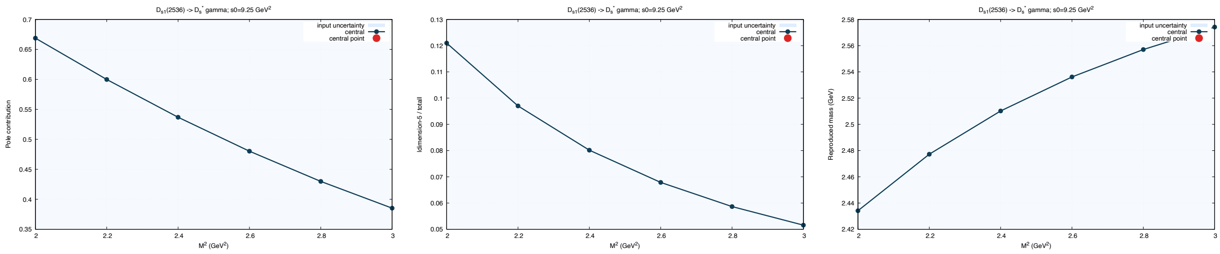
Figure 2: The four principal stability scans for the higher bottom state.



(a) Charm lower-state angle scan.

(b) Bottom higher-state angle scan.

Figure 3: Mixing-angle sensitivity. Zero crossings expose destructive interference between the A and B current components.



(a) pole contribution

(b) dimension-five fraction

(c) mass reconstruction

Figure 4: Representative two-point diagnostics for $D_{s1}(2536)$.

6.3 Comparison with previous calculations and data

Table 6: Radiative-width predictions and current experimental status. Width entries are in keV. "This work" gives the median and 16th–84th percentile interval; experimental rows report direct branching-fraction, total-width, or observation information and must not be read as radiative widths.

Entry	width [keV] or experimental status	method/status	source and note
$D_{s1}(2460)^+ \to D_s^{*+}\gamma$			
This work	$0.482^{+1.609}_{-0.434}$	mixed-current external-photon LCSR	this calculation
Published	0.6 – 1.1	pure-axial external-photon LCSR	Colangelo et al. (2005) [3]; pure-A benchmark
Published	1.5	vector-meson dominance	Colangelo et al. (2005) [3]
Published	5.5	nonrelativistic quark model	Godfrey (2003) [5]
Published	4.66	chiral-doublet effective theory	Bardeen et al. (2003) [6]
Published	4.74 – 4.79	relativistic quark model	Chen et al. (2020) [7]; three parameter modes
Published	15.5	potential-model calculation	Radford et al. (2009) [8]
Published	17.4	potential-model calculation	Green et al. (2017) [9]
Published	5.6	relativized quark model	Godfrey (2005) [10]
Published	4.41	quark model	Close and Swanson (2005) [11]
Published	14.6 – 22.8	relativistic heavy-quark model	Goity and Roberts (2001) [12]; three model variants collected in Chen et al.
Published	13 ± 2	hadronic-molecule EFT	Fu et al. (2022) [13]; full result
Published	104	potential model with relativistic corrections	Bondar and Milstein (2025) [14]; authors estimate 50% model uncertainty
Experiment	$\mathcal{B}(D_s^*\gamma) < 8\%$ **(90% CL); $\Gamma_{\text{tot}} < 3.5$ MeV**	upper limit	PDG (2026) [1]
$D_{s1}(2536)^+ \to D_s^{*+}\gamma$			
This work	$0.094^{+0.520}_{-0.089}$	mixed-current external-photon LCSR	this calculation
Published	2.96 – 3.02	relativistic quark model	Chen et al. (2020) [7]; three parameter modes
Published	8.90	potential-model calculation	Radford et al. (2009) [8]
Published	9.21	potential-model calculation	Green et al. (2017) [9]
Published	0.4 ± 1.0	heavy-quark effective calculation	Körner et al. (1993) [15]
Published	5.5	relativized quark model	Godfrey (2005) [10]
Published	1.59	quark model	Close and Swanson (2005) [11]
Published	14.0 – 25.1	relativistic heavy-quark model	Goity and Roberts (2001) [12]; three model variants collected in Chen et al.
Published	29	potential model with relativistic corrections	Bondar and Milstein (2025) [14]; authors estimate 50% model uncertainty
Experiment	**mode possibly seen; no $\mathcal{B}$ or partial width; $\Gamma_{\text{tot}} = 0.92 \pm 0.05$ MeV**	possibly seen	PDG (2026) [1]

Continued on the next page

Table 6 continued from the previous page

Entry	width [keV] or experimental status	method/status	source and note
$B_{s1}^{L\,0} \to B_s^{*0}\gamma$ **(predicted lower state)**			
This work	$1.589^{+2.703}_{-1.298}$	mixed-current external-photon LCSR	this calculation
Published	0.3 – 6.1	pure-axial external-photon LCSR	Wang (2008) [4]; lower strange-bottom axial partner
Published	100 ± 15	hadronic-molecule EFT	Fu et al. (2022) [13]; lower strange-bottom molecular partner
Published	60	nonrelativistic quark model	Li et al. (2021) [16]; conditional subthreshold missing 1P1 state
Published	20	potential model with relativistic corrections	Bondar and Milstein (2025) [14]; $B_{s1}(5748)$ assignment; authors estimate 50% accuracy
Experiment	**state unobserved; no measurement or limit**	unobserved state	PDG (2026) [1]
$B_{s1}(5830)^0 \to B_s^{*0}\gamma$ **(observed higher state)**			
This work	$2.332^{+5.735}_{-1.925}$	mixed-current external-photon LCSR	this calculation
Published	53	nonrelativistic quark model	Li et al. (2021) [16]; observed higher state
Published	41	potential model with relativistic corrections	Bondar and Milstein (2025) [14]; $B_{s1}(5829)$ assignment; authors estimate 50% accuracy
Experiment	**radiative mode unobserved; no measurement or limit**	unobserved mode	PDG (2026) [1]

Table 6 includes our four percentile results, the predictions in the primary comparison tables of Refs. [3, 7, 13], and the additional bottom-state calculation of Ref. [16]. We also include the relativistic-potential model of Ref. [14], which predicts 104 and 29 keV for the two charm channels and 20 and 41 keV for its $B_{s1}(5748)$ and $B_{s1}(5829)$ assignments, respectively; that work estimates a 50% model uncertainty for these predictions. Values quoted for several parameter modes or model variants are compressed to their displayed range; no uncertainty has been inferred where the source gives only a central value. The lower B_{s1}^L rows of Refs. [4, 13, 14, 16] concern predicted near-threshold partners and are not assignments of the observed $B_{s1}(5830)$.

Experiment does not yet provide a radiative partial width for any of the four channels. For $D_{s1}(2460)^+ \to D_s^{*+}\gamma$, the direct result is the 90% C.L. branching-fraction bound $\mathcal{B}(D_s^*\gamma) < 8\%$, accompanied by the independent total-width bound $\Gamma_{\rm tot} < 3.5$ MeV; these are therefore shown as limits rather than converted into a measured width. The $D_{s1}(2536)$ mode is listed as possibly seen without a branching fraction, while the lower B_{s1}^L is unobserved and no quantitative $B_{s1}(5830) \to B_s^*\gamma$ limit is available [1].

7 Independent validation

The symbolic Wolfram Language calculation constructs explicit Dirac matrices, evaluates AA, AB, and BB cut traces, factorizes equation (13), rotates the matrix, and checks the Ward determinant. The numerical Wolfram Language implementation uses native adaptive integration and does not read Python values. Python uses Gauss–Legendre quadrature. Their complete 42-entry comparison is machine-readable in `outputs/csv/python_mathematica_regression.csv`. All entries pass either 2×10^{-8} absolute or 2×10^{-5} relative tolerance. The maximum material relative difference is 9.13e-07; the maximum absolute difference is 2.43e-06.

Gate	status	evidence
$\rho_{AB} = \rho_{BA}$	checked	identical function and symbolic WL equality
spectral positivity	checked	analytic determinant is nonnegative above threshold
mass dimensions	checked	rho dim 2; Borel matrix and T dim 4; g dimensionless
physical threshold	checked	all dispersive integrals start at (mQ+ms)^2
Ward identity	checked	epsilon structure vanishes under epsilon_gamma -> q
no local transition condensate	checked	diagnostic term excluded from quoted_total_A/B
two-point $d = 3, 5$	checked	explicit matrices retained
pure A limit	checked	theta=90 degrees for lower state
pure B limit	checked	theta=0 degrees for lower state
equal Borel limit	checked	M1^2=M2^2=2M^2 gives u0=1/2
$m_s \to 0$ threshold	checked	lambda and density reduce smoothly
charge switch	checked	hard density changes through eQ/es only
Python–Mathematica regression	checked	42/42 entries pass; max material rel=9.128e-07; max abs=2.425e-06
subleading tensor kernels	truncation	not retained; 15% charm and 5% bottom correlated uncertainty assigned
transition α_s	truncation	not retained

8 Conclusions

The clean-room calculation produces a fully populated two-point current matrix, mixed-state residues, a term-resolved external-photon LCSR, four radiative couplings and widths, stability scans, uncertainty samples, and an independent Wolfram Language regression. The median widths are 0.482 keV for $D_{s1}(2460)$, 0.094 keV for $D_{s1}(2536)$, 1.589 keV for the lower B^L_{s1}, and 2.332 keV for $B_{s1}(5830)$, with the percentile ranges in table 2.

The dominant conclusion is methodological: mixed-current interference is numerically important, especially for the higher states, and the transition Borel dependence prevents precision claims. A next-order study should derive the subleading tensor-current kernels independently for every DA, include α_s corrections, revisit scale running consistently, and calculate the second on-shell multipole.

A Photon distribution amplitudes

At $\mu = 1$ GeV the two-particle shapes used in the numerical code are

$$\phi_\gamma(u) = 6u\bar{u}, \tag{52}$$

$$h_\gamma(u) = -10C_2^{1/2}(2u-1), \tag{53}$$

$$\psi^v(u) = 5[3\xi^2 - 1] + \frac{3}{64}(15\omega_\gamma^V - 5\omega_\gamma^A)[3 - 30\xi^2 + 35\xi^4], \tag{54}$$

$$\psi^a(u) = \frac{5}{2}(1-\xi^2)(5\xi^2-1)\left(1 + \frac{9}{16}\omega_\gamma^V - \frac{3}{16}\omega_\gamma^A\right), \tag{55}$$

$$\begin{aligned}\mathcal{A}(u) = {} & 40u^2\bar{u}^2(3\kappa+1) - 24\zeta_2\big[u\bar{u}(2+13u\bar{u}) \\ & + 2u^3(10-15u+6u^2)\ln u + 2\bar{u}^3(10-15\bar{u}+6\bar{u}^2)\ln\bar{u}\big],\end{aligned} \tag{56}$$

where $\bar{u} = 1-u$ and $\xi = 2u-1$.

With $\alpha_q + \alpha_{\bar{q}} + \alpha_g = 1$, the three-particle DAs are

$$\mathcal{V} = 540\omega_\gamma^V(\alpha_q - \alpha_{\bar{q}})\alpha_q\alpha_{\bar{q}}\alpha_g^2, \tag{57}$$

$$\mathcal{A}_3 = 360\alpha_q\alpha_{\bar q}\alpha_g^2\left[1+\frac{\omega_\gamma^A}{2}(7\alpha_g-3)\right], \tag{58}$$

$$\mathcal{S} = 30\alpha_g^2\{\kappa(1-\alpha_g)+\zeta_1(1-\alpha_g)(1-2\alpha_g)+\zeta_2[3(\alpha_{\bar q}-\alpha_q)^2-\alpha_g(1-\alpha_g)]\}, \tag{59}$$

$$\widetilde{\mathcal{S}} = -\mathcal{S}, \tag{60}$$

$$\mathcal{T}_1 = \mathcal{T}_3 = -360\zeta_2(\alpha_{\bar q}-\alpha_q)\alpha_q\alpha_{\bar q}\alpha_g, \tag{61}$$

$$\mathcal{T}_2 = \mathcal{T}_4 = 30\alpha_g^2(\alpha_{\bar q}-\alpha_q)\{\kappa+\zeta_1(1-2\alpha_g)+\zeta_2(3-4\alpha_g)\}. \tag{62}$$

The combinations are

$$\mathcal{F}_2 = \mathcal{S}+\widetilde{\mathcal{S}}+\mathcal{T}_1-\mathcal{T}_2-\mathcal{T}_3+\mathcal{T}_4, \qquad \mathcal{F}_3 = \mathcal{A}_3+\mathcal{V}. \tag{63}$$

Their post-Borel integrals are

$$I_{\mathcal{F}_2} = \int_0^{1-u_0} dv \int_0^{u_0/(1-v)} d\alpha_g\,\mathcal{F}_2 + \int_{1-u_0}^{1} dv \int_0^{(1-u_0)/v} d\alpha_g\,\mathcal{F}_2, \tag{64}$$

$$\alpha_q = u_0-(1-v)\alpha_g, \quad \alpha_{\bar q} = 1-u_0-v\alpha_g, \tag{65}$$

$$I_{\mathcal{F}_3} = \int_0^{u_0} d\alpha_{\bar q}\left[\int_{u_0-\alpha_{\bar q}}^{1-\alpha_{\bar q}} \frac{d\alpha_g}{\alpha_g^2}\mathcal{F}_3(1-\alpha_{\bar q}-\alpha_g,\alpha_{\bar q},\alpha_g) - \frac{\mathcal{F}_3(1-u_0,\alpha_{\bar q},u_0-\alpha_{\bar q})}{u_0-\alpha_{\bar q}}\right]. \tag{66}$$

For the central symmetric BBK parameters at $u_0 = 1/2$, $I_{\mathcal{F}_2} = 0$ and $I_{\mathcal{F}_3} = 15(\omega_\gamma^A - 3\omega_\gamma^V + 4)/32 = -285/64$.

B Supplementary figure index

For every state the project contains PDF and PNG versions of g and Γ versus M^2 and s_0, g versus angle, and two-point pole contribution, reconstructed mass, residue, and dimension-five fraction versus M^2. Each plot has a companion CSV. This paper shows representative panels; the detailed derivation report embeds the complete set.

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